

Mohammed Ansar Ur Rahman

+1 (346) 699-6777 ansarurrahman5@outlook.com — Houston, Texas
linkedin.com/in/mohammed-ansar-ur-rahman — github.com/ansar534

Education

University of Houston, Houston, TX May 2027
Master of Science, Engineering Data Science & AI
Relevant Coursework: Probability & Statistics, Applied Neural Networks, AI for Engineers, Database Management

Anurag University, Hyderabad, India Nov 2021 – Apr 2025
Bachelor of Technology, Computer Science and Engineering (Data Science), GPA: 3.63/4.0
Relevant Coursework: Data Structures & Algorithms, Deep Learning, Big Data Analytics, Statistical Methods for Data Analysis

Technical Skills

AI/ML & Programming: LLMs, RAG, Agentic AI, LangChain, MCP, Semantic Search, Vector Databases, NLP, Scikit-learn, Python, SQL, Pandas
Visualization & Databases: Power BI, Tableau, Excel, MySQL, PostgreSQL, Snowflake
Cloud & DevOps: Azure, AWS, Docker, Kubernetes, FastAPI, Git, GitHub Actions

Experience

GenAI Intern April 2026 – Present
NOV Inc, Houston, TX

- Design an enterprise-scale semantic search and retrieval platform using vector embeddings and cosine similarity matching over tens of thousands of unstructured support and documentation records, enabling intent-aware search across the organization
- Design and build a Model Context Protocol (MCP) server exposing 20 structured AI tools spanning search, resolution recommendation, analytics, and vector store management, integrated with n8n for no-code workflow automation
- Develop an agentic orchestration pipeline with a tool-calling planner and LLM synthesis layer, adding confidence scoring and guardrails to reduce hallucinated responses
- Build automated data ingestion pipelines (Python, REST APIs) with incremental sync, deduplication, and audit logging, containerized with Docker and deployed as scheduled Kubernetes CronJobs on Azure Kubernetes Service across Dev/QA/Prod via a GitHub Actions CI/CD pipeline

Research Assistant Sep 2025 – April 2026
C.T. Bauer College of Business, University of Houston, Houston, TX

- Built machine learning models on 4,000+ student records, improving academic readiness prediction accuracy by 25% using Python, Pandas, and Scikit-learn
- Developed SQL and Excel KPI dashboards, increasing training effectiveness by 15% and reducing manual processing time by 40%

Data Analyst Oct 2025 – May 2026
University of Houston – Muslim Student Association (MSA), Houston, TX

- Built interactive Power BI dashboards for 500+ members, improving outreach effectiveness by 20% and event planning efficiency by 30%
- Automated statistical reporting pipelines, reducing manual report generation time by 60% and enhancing data storytelling

Python Developer Intern May 2024 – June 2024
Cognifyz Technologies, Remote

- Built Python REST APIs with input validation and structured logging, improving system reliability and performance by 20%
- Containerized services using Docker to enable scalable ETL workflows and streamlined deployments

Projects

Retrieval-Augmented Generation (RAG) Agent — Python, LangChain, ChromaDB, Ollama 2025
• Designed an end-to-end RAG system that ingests PDFs and CSVs, generates vector embeddings, and answers user queries using a local LLM
• Optimized local LLM inference via Ollama by tuning model selection and context size for low-memory (4GB RAM) environments

Legal AI — Contract & Compliance Intelligence Platform | Python, LangChain, ChromaDB, Groq 2025
API, HuggingFace, Streamlit
• Built a RAG-based legal intelligence platform covering contract Q&A, risk review, litigation assistance, and compliance auditing over regulatory documents
• Implemented semantic retrieval using ChromaDB and HuggingFace sentence-transformers, with Groq-hosted LLM inference for low-latency query responses

Coal Production Forecasting System — Python, ARIMA, SARIMA, FastAPI, Streamlit 2025
• Developed time-series forecasting models across 10+ mining sites, achieving 85% prediction accuracy and reducing forecast variance by 30%

Publications

AI-Driven Renewable Energy Consumption Prediction and Optimization System, *International Journal of Applied Engineering Research*, 2025